

CSE 565 Design of Experiments Project

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CSE 565: Software Verification and Validation

Design of Experiments Project

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1 Introduction and Mobile Application Specification

This project applies pairwise Design of Experiments (DOE) testing to a mobile application that collects five customer inputs. Pairwise testing reduces the full input space while ensuring that every possible pair of factor values occurs in at least one test case. The complete exhaustive space contains $5 \times 3 \times 4 \times 4 \times 5 = 1,200$ combinations.

Table 1: Mobile application factors and levels.

Factor	Allowed levels
Type of Phone	iPhone 14; iPhone 13; Galaxy Z; Huawei Mate; Google Pixel 7
Authentication	Fingerprint; Face recognition; Text Password
Connectivity	Wireless; 3G; 4G LTE; 5G Edge
Memory	128 GB; 256 GB; 512 GB; 1 TB
Battery Level	< 20%; 20–39%; 40–59%; 60–79%; 80–100%

2 Task 1: Pairwise Test Cases Created by a DOE Tool

2.1 DOE Tool Selection

I selected Microsoft PICT (Pairwise Independent Combinatorial Testing) as the DOE tool. PICT accepts a plain-text model of factors and levels and generates a compact pairwise test suite. I used PICT version 3.7.4 with its default pairwise strength (two-way coverage).

2.2 PICT Model

The factor model used by PICT is stored in `pict_model.txt`. It contains the five factors and all permitted levels from the assignment specification.

```
# CSE 565 Design of Experiments Project
# Pairwise (2-way) input model for the specified mobile application.

Type of Phone: iPhone 14, iPhone 13, Galaxy Z, Huawei Mate, Google Pixel 7
Authentication: Fingerprint, Face recognition, Text Password
Connectivity: Wireless, 3G, 4G LTE, 5G Edge
Memory: 128 GB, 256 GB, 512 GB, 1 TB
Battery Level: <20%, 20-39%, 40-59%, 60-79%, 80-100%
```

2.3 PICT Generation and Results

The following command generated the pairwise test suite:

```
pict pict_model.txt
```

PICT reported 175 required value-pairs and generated 26 test cases. The raw output is stored in `pict_pairwise_test_cases.tsv`.

Screenshot placeholder

Replace with a screenshot of the PICT command, model, and generated output.

3 Task 2: Pairwise Test Cases Created by a Generative AI Tool

3.1 Generative AI Tool Selection

ChatGPT will be used to independently generate a second pairwise test suite. The prompt will provide the same factor/level specification used by PICT and will request a complete table containing only allowed values.

3.2 Prompt Used

Create a pairwise (2-way) test suite for a mobile application. Return a complete tab-separated table with exactly these five columns: Type of Phone, Authentication, Connectivity, Memory, and Battery Level.

Use only these levels:

- Type of Phone: iPhone 14, iPhone 13, Galaxy Z, Huawei Mate, Google Pixel 7
- Authentication: Fingerprint, Face recognition, Text Password
- Connectivity: Wireless, 3G, 4G LTE, 5G Edge
- Memory: 128 GB, 256 GB, 512 GB, 1 TB
- Battery Level: <20%, 20-39%, 40-59%, 60-79%, 80-100%

Every pair of values from every two different columns must occur in at least one row. Do not use values outside these lists. Do not include duplicate rows. After the table, state the number of test cases generated.

3.3 Generated Test Cases and Evidence

ChatGPT generated 27 test cases in tab-separated format. The exact prompt and unedited response are preserved in `chatgpt_prompt_and_response.md`, and the table used for validation is stored in `chatgpt_pairwise_test`

Screenshot placeholder

Replace with a screenshot showing the exact ChatGPT prompt and generated test-case table.

4 Task 3: Assessment of the Two Test Suites

Both suites will be independently validated rather than relying only on the generation tool’s coverage claim. The verification checks that every test case is complete, every value is permitted by the specification, duplicate rows are identified, and every required pair is covered.

Table 2: Pairwise-suite comparison. Complete this table after generating and verifying the AI suite.

Measure	PICT suite	ChatGPT suite
Test cases generated	26	27
Invalid values	0	0
Duplicate rows	0	0
Required value-pairs	175	175
Pairs covered	175/175	175/175
Pairwise coverage	100%	100%

Both suites were checked with `verify_pairwise_coverage.py`. The PICT suite contains 26 valid, non-duplicate test cases and covers all 175 required value-pairs. The ChatGPT suite contains 27 valid, non-duplicate test cases and also covers all 175 required value-pairs. The PICT suite is one test case smaller and is reproducible from the same model and command. The ChatGPT result achieved the same coverage, but required a precise prompt and independent verification.

The two suites therefore satisfy the pairwise DOE requirement. Their results show that generative AI can produce a valid pairwise suite for this small model, but its stated coverage should not be accepted without validation. A deterministic DOE tool provides a more repeatable generation workflow, while an AI tool can assist with quickly creating and explaining a candidate suite.

Screenshot placeholder

Replace with a screenshot of the independent verification output for both suites.

5 Task 4: Assessment of the DOE Tool

5.1 Features and Functionality

5.2 Scope Covered by the Tool

5.3 Performance

5.4 Ease of Use

6 Task 5: Assessment of the Generative AI Tool

6.1 Prompting and Result-Processing Experience

6.2 Significance of Generative AI for DOE Testing

7 Conclusion

References

Microsoft. (n.d.). *PICT: Pairwise Independent Combinatorial Testing*. GitHub. Retrieved September 21, 2026, from <https://github.com/microsoft/pict>